Ex-7

24/09/2026

SLIDING WINDOW PROTOCOL

AIM:-

→ Write a program to implement flow ~~data~~ control at data link layer using Sliding Window Protocol. Simulate the flow of frames from one node to another.

SENDER PROGRAM:-

```
import time
w = int(input("Window size: "))
msg = input("Message: ")
frames = [(i, c) for i, c in enumerate(msg)]
i = 0
while i < len(frames):
    end = min(i+w, len(frames))
    with open("Sender-Buffer.txt", "w") as f:
        for n, c in frames[i:end]:
            print("Frame", n, ":", c)
            f.write(f"{n}, {c}\n")
```

```
time.sleep(2)
with open("Receiver-Buffer.txt") as f:
    ack = f.read().strip()
if ack.startswith("ACK:"):
    a = int(ack[4:])
    if a == end:
        i = end
    else:
        print("Wrong ACK - Resending")
elif ack.startswith("NACK:"):
    i = int(ack[5:])
    print("NACK - Resending from", i)
print("Transmission complete")
```

RECEIVER PROGRAM:-

```
import time
import os
expected = 0
while True:
    if not os.path.exists("Sender_Buffer.txt"):
        time.sleep(1)
        continue
    with open("Send_Buffer.txt") as f:
        frames = f.readlines()
    if not frames:
        time.sleep(1)
        continue
    error = False
    for line in frames:
        n, data = line.strip().split(",")
        n = int(n)
        print("Received:", n, data)
```

```
        if n!=expected:
            with open("Receiver_Buffer.txt", "w") as f:
                f.write("NACK:" + str(expected))
            error: True
            break
        expected += 1
    if not error:
        with open("Receiver_Buffer.txt", "w") as f:
            f.write("ACK:" + str(expected))
        error = True
        break
    expected += 1
if not error:
    with open("Receiver_Buffer.txt", "w") as f:
        f.write("ACK:" + str(expected))
open("Sender_Buffer.txt", "w").close()
time.sleep(1)
```

SENDER CODE INPUT:-

Window size: 3

Message: HELLO

SENDER CODE OUTPUT:-

Frame 0 : H

Frame 1 : E

Frame 2 : L

Frame 3 : L

Frame 4 : O

~~Recv~~ RECEIVER CODE INPUT:-

Sender_Buffer.txt

0, H

1, E

2, L

3, L

4, O



RECEIVER CODE OUTPUT:-

Received: 0 H

Received: 1 E

Received: 2 L

Received: 3 L

Received: 4 O

RESULT:-

The sliding Window Protocol was successfully implemented to control frame flow and handle acknowledgements and retransmissions.